BME i9400 – Capstone Project

Fall 2026

The capstone is an individual study. You choose one of four datasets, each with a fixed supervised learning task, and you build the best model you can for that task. "Best" is measured on a held-out test set that you use once, at the end, and the improvement over a provided baseline must be shown with a confidence interval. What is graded is the quality of the study: how the model was searched for, how it was selected, how it was evaluated, and how well you can explain and defend it. A strong model with a weak protocol scores lower than a modest model with a sound one.

The capstone is 35% of the course grade: the oral presentation and live Q&A 20%, the written report 15%.

1 – THE FOUR DATASETS

Each dataset comes with a starter notebook that loads the data, applies the fixed split, fits the baseline, and reports the primary metric. Everything runs on a free Colab CPU.

	DATASET	MODALITY	THE TASK	PRIMARY METRIC
A	Diabetes 130-US Hospitals	CLINICAL TABULAR	30-day hospital readmission	AUROC
B	PathMNIST	HISTOLOGY IMAGES	Tissue type in colorectal cancer, 9 classes	Balanced accuracy
C	MIT-BIH Arrhythmia	ECG	Heartbeat class, 4 classes, unseen patients	Macro-averaged F1
D	PubMed 20k RCT	BIOMEDICAL TEXT	Role of each sentence in an abstract, 5 classes	Macro-averaged F1

> **Balanced accuracy** is the mean of the per-class sensitivities. Unlike accuracy, it is not dominated by the most common class.
>
> **Macro-averaged F1** computes the F1 score (the harmonic mean of precision and sensitivity) for each class separately and averages them with equal weight, so a rare class counts as much as a common one.

A – Readmission after a diabetes hospitalisation

- **Data.** About 100,000 hospital encounters of patients with diabetes, 130 US hospitals, 1999–2008; Strack et al., *BioMed Research International*, 2014; UCI Machine Learning Repository, CC BY 4.0.
- **A row of X.** One encounter: demographics, admission type and source, length of stay, numbers of procedures, medications and diagnoses, three ICD-9 diagnosis codes, laboratory results, and diabetes medication changes.
- **y.** 1 if the patient was readmitted within 30 days of discharge, 0 otherwise (about 11% positive).
- **The split.** Fixed in the starter notebook, **by patient**: every encounter of a patient is in the same set.
- **Baseline.** L2-penalised logistic regression on a fixed set of features.
- **Directions.** Gradient boosting (Meeting 19); grouping the ICD-9 codes into clinical categories; features from a patient's earlier encounters; class imbalance.
- **Known pitfalls.** Many patients have several encounters, so a split by encounter leaks information between the training and test sets. Encounters that end in death or discharge to hospice cannot be followed by a readmission. Missing values are recorded as `?`, and `weight` is missing for almost every encounter.

B – Tissue classification in colorectal cancer histology

- **Data.** PathMNIST, from the MedMNIST v2 collection (Yang et al., *Scientific Data*, 2023), built from the colorectal cancer histology datasets of Kather et al. (*PLOS Medicine*, 2019): 107,180 patches of H&E-stained tissue, resized to 28 × 28 pixels. CC BY 4.0.
- **A row of X.** One RGB image patch, 28 × 28 × 3.
- **y.** One of 9 tissue classes: adipose, background, debris, lymphocytes, mucus, smooth muscle, normal colon mucosa, cancer-associated stroma, colorectal adenocarcinoma epithelium.

- **The split.** The official MedMNIST training, validation and test sets (89,996 / 10,004 / 7,180). The training and validation patches come from the same 86 slides; the test patches come from 50 other patients.
- **Baseline.** Multinomial logistic regression on the pixel values, averaged over 2×2 blocks.
- **Directions.** A small CNN; a pretrained image encoder used frozen or fine-tuned (Meetings 21–22); data augmentation (flips, rotations, colour jitter for staining differences); Grad-CAM for the error analysis.
- **Known pitfalls.** Validation scores overestimate test scores, because validation patches share slides with the training patches while the test patients are new. The class proportions of the test set differ from those of the training set. Training a CNN on all 89,996 patches takes one to two minutes per pass on a CPU.

C — Heartbeat classification in patients the model has not seen

- **Data.** MIT-BIH Arrhythmia Database (Moody and Mark, *IEEE Engineering in Medicine and Biology*, 2001), PhysioNet: 48 half-hour two-lead ECG recordings from 47 patients, 360 samples per second, with every beat annotated by cardiologists. Open Data Commons Attribution License.
- **A row of X.** One heartbeat: a window of the lead II signal centred on the annotated R peak, with the intervals to the previous and next beats.
- **y.** The AAMI beat class: N (normal and bundle-branch block), S (supraventricular ectopic), V (ventricular ectopic), F (fusion). The fifth class, Q (unclassifiable, 15 beats), is excluded.
- **The split.** The **inter-patient** split of de Chazal et al. (*IEEE Transactions on Biomedical Engineering*, 2004): 22 recordings for development, 22 different recordings for testing, the four recordings with paced beats excluded.
- **Baseline.** Logistic regression on the RR intervals and a down-sampled beat waveform.
- **Directions.** A 1D CNN on the waveform (Meeting 23); combining learned waveform features with RR-interval features; class imbalance; normalising each patient's RR intervals.
- **Known pitfalls.** A random split of beats places beats of the same patient in both the training and test sets and gives accuracies near 99%; many published results are of this kind. Under the inter-patient split the S class is hard, and its sensitivity is the number to watch.

D — The role of a sentence in a medical abstract

- **Data.** PubMed 20k RCT (Dernoncourt and Lee, *IJCNLP*, 2017): about 20,000 abstracts of randomised controlled trials, about 240,000 sentences.
- **A row of X.** One sentence, with its position in the abstract and the abstract it belongs to.
- **y.** The role of the sentence: background, objective, methods, results, or conclusions.
- **The split.** The official training, development and test sets, which are divided by abstract.
- **Baseline.** TF-IDF features of the sentence and multinomial logistic regression.
- **Directions.** The position of the sentence in the abstract; features of the neighbouring sentences; frozen sentence embeddings from a small pretrained encoder (Meeting 24) with a classifier on top.
- **Known pitfalls.** Fine-tuning a transformer on all 180,000 training sentences is not feasible on a CPU; the starter notebook shows how to subsample, and the subsample size must be reported. Objective and background sentences are easily confused.

Your own data

You may petition to use data from your own research instead. Submit the petition with the analysis plan on **Monday, November 23**. It must state:

1. the question, what a row of X is, and what y is
2. the number of examples, and the number in each class (or the distribution of y)
3. where the data come from, and written permission from the data owner (usually your advisor)
4. that the data are de-identified, and will not be uploaded to any AI service if they contain anything that could identify a person
5. how the data will be split, including what keeps related examples (the same patient, subject or session) in the same set
6. the baseline you will fit

A petition is approved if the task is supervised, the data are large enough for a held-out test set, and the study can be run in Colab. The requirements of sections 2 to 4 are unchanged.

2 – WHAT THE STUDY MUST CONTAIN

1. **The task.** The intended use of the model, what *X* and *y* are, where the data come from, and the fixed split.
2. **The baseline, reproduced.** Run the starter notebook and report its result on the validation data.
3. **The model search.** At least two model families beyond the baseline. For each, state the hyperparameters searched and their ranges, and how the choice was made. Every selection decision — model family, hyperparameters, features, threshold — is made on validation data or by cross-validation on the development set.
4. **The test set, once.** The final model and the baseline are evaluated on the test set once, at the end. Report the primary metric for both, and a **95% bootstrap confidence interval for the difference** between them (Meeting 20), resampling the independent units: patients (A), image patches (B), recordings (C), abstracts (D). Each starter notebook contains a function that does this.
5. **Error analysis.** Where the final model fails: the confusion matrix, and at least one of a subgroup analysis (age, sex, hospital, patient, abstract length), the hardest examples, or a saliency map (dataset B).
6. **AI use log.** The tools used, what for, and any output that materially shaped the work.

> **The test labels are in the data you are given.** Using them for any decision, including choosing between two models after looking at both of their test scores, makes the test result an overestimate. Report validation results for every model you tried; report test results for the final model and the baseline only.

3 – DELIVERABLES AND DATES

DATE	MILESTONE
Mon Nov 16	Briefs and starter notebooks released
Mon Nov 23	**Dataset choice and analysis plan**, one page, submitted in class: the dataset (or petition), the primary metric, the model families you intend to try, and the baseline reproduced
Mon Dec 7	Capstone clinic: bring your validation results table; peer critique
Wed Dec 9, Mon Dec 14, exam week	Presentations: 8 minutes and 4 minutes of graded Q&A. The order is drawn at random
Exam week, date to be announced	Written package: the notebook, the report, the AI use log

Capstone milestones lose 20% per day late (syllabus).

The report includes figures, under the headings: Introduction, Data and Methods, Results, Discussion. The AI use log should be added as an appendix.

Submit everything by pull request into `capstone/submissions/`.

4 – HOW IT IS GRADED

Presentation and Q&A (20% of the course grade).

Graded for clarity of presentation, soundness of the presented approach, interpretation of the findings, and how accurately the questions are answered.

Written report (15% of the course grade).

ITEM	WEIGHT
Task and protocol: intended use, X and y, the split and why it is sound	15%
Baseline reproduced and model search: families, hyperparameter spaces, selection on validation data	30%
Evaluation: test results used once, confidence interval for the difference from the baseline	25%
Error analysis and limitations	20%
Reproducibility and clarity	10%

The size of the improvement over the baseline is not graded directly. A model that does not beat the baseline, evaluated honestly and explained well, can receive full marks.

Generative AI is permitted and expected. You are responsible for every line and every claim, and the Q&A tests whether you can account for them without assistance.